betaDelta: Internal Tests

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Tests

```
#> test-betaDelta-beta-delta-adf
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-beta-delta-methods

#> Call:
#> BetaDelta(object = object, type = "mvn")
#>
#> Standardized regression slopes with MVN standard errors:
#>            est     se      t df     p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0759 6.5272 42 0.000  0.2268 0.2905 0.3421 0.6482 0.6998
#> PCTGRT  0.3915 0.0770 5.0824 42 0.000  0.1190 0.1837 0.2360 0.5469 0.5993
#> PCTSUPP 0.2632 0.0747 3.5224 42 0.001 -0.0011 0.0616 0.1124 0.4141 0.4649
#>         99.95%
#> NARTIC  0.7635
#> PCTGRT  0.6640
#> PCTSUPP 0.5276
#> Call:
#> BetaDelta(object = object, type = "mvn")
#>
#> Standardized regression slopes with MVN standard errors:
#>            est     se      t df     p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0759 6.5272 42 0.000  0.2268 0.2905 0.3421 0.6482 0.6998
#> PCTGRT  0.3915 0.0770 5.0824 42 0.000  0.1190 0.1837 0.2360 0.5469 0.5993
#> PCTSUPP 0.2632 0.0747 3.5224 42 0.001 -0.0011 0.0616 0.1124 0.4141 0.4649
#>         99.95%
#> NARTIC  0.7635
#> PCTGRT  0.6640
#> PCTSUPP 0.5276
#> Call:
#> BetaDelta(object = object, type = "adf")
#>
#> Standardized regression slopes with ADF standard errors:
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
```

```
#> NARTIC  0.4951 0.0674 7.3490 42 0.0000  0.2568 0.3134 0.3592 0.6311 0.6769
#> PCTGRT  0.3915 0.0710 5.5164 42 0.0000  0.1404 0.2000 0.2483 0.5347 0.5830
#> PCTSUPP 0.2632 0.0769 3.4231 42 0.0014 -0.0088 0.0558 0.1081 0.4184 0.4707
#>         99.95%
#> NARTIC  0.7335
#> PCTGRT  0.6426
#> PCTSUPP 0.5353
#> Call:
#> BetaDelta(object = object, type = "adf")
#>
#> Standardized regression slopes with ADF standard errors:
#>            est     se      t df      p   0.05%   0.5%   2.5%  97.5%  99.5%
#> NARTIC  0.4951 0.0674 7.3490 42 0.0000  0.2568 0.3134 0.3592 0.6311 0.6769
#> PCTGRT  0.3915 0.0710 5.5164 42 0.0000  0.1404 0.2000 0.2483 0.5347 0.5830
#> PCTSUPP 0.2632 0.0769 3.4231 42 0.0014 -0.0088 0.0558 0.1081 0.4184 0.4707
#>         99.95%
#> NARTIC  0.7335
#> PCTGRT  0.6426
#> PCTSUPP 0.5353
#> Call:
#> BetaDelta(object = object, type = "mvn")
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0618 12.3341 44 0 0.5443 0.5958 0.6376 0.8867 0.9285   0.98
#> Call:
#> BetaDelta(object = object, type = "mvn")
#>
#> Standardized regression slopes with MVN standard errors:
#>           est     se       t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0618 12.3341 44 0 0.5443 0.5958 0.6376 0.8867 0.9285   0.98
#> Call:
#> BetaDelta(object = object, type = "adf")
#>
#> Standardized regression slopes with ADF standard errors:
#>           est     se      t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0604 12.625 44 0 0.5493 0.5996 0.6405 0.8838 0.9247  0.975
#> Call:
#> BetaDelta(object = object, type = "adf")
#>
#> Standardized regression slopes with ADF standard errors:
#>           est     se      t df p  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0604 12.625 44 0 0.5493 0.5996 0.6405 0.8838 0.9247  0.975
#> Test passed with 1 success.

#> test-betaDelta-beta-delta-mvn
```

```
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-beta-delta-std

#> Test passed with 10 successes.

#> test-betaDelta-beta-star-select

#> Test passed with 8 successes.

#> test-betaDelta-delta-r-sq-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 3 successes.

#> test-betaDelta-diff-beta-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaDelta-es-delta-methods

#> Test passed with 181 successes.

#> test-betaDelta-f-sq-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 3 successes.

#> test-betaDelta-jac-beta-star-sel-wrt-theta-star

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaDelta-jac-delta-r-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-jac-f-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.
```

```
#> test-betaDelta-jac-p-cor-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-jac-r-sq-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-jac-s-cor-wrt-theta-star

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 2 successes.

#> test-betaDelta-jac-theta-star-wrt-vech-sigma

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaDelta-jac-vech-sigma-wrt-theta-star

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaDelta-p-cor-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 3 successes.

#> test-betaDelta-r-sq-delta-se

#> Test passed with 1 success.

#> test-betaDelta-r-sq-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 3 successes.

#> test-betaDelta-s-cor-delta

#> Test passed with 2 successes.
#> Test passed with 2 successes.
#> Test passed with 3 successes.
#> Test passed with 1 success.
```

```
#> test-deltaMethod-delta-generic

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> DeltaGeneric(object = object, def = def, alpha = 0.05)
#>              est     se       z      p    2.5%   97.5%
#> exp(wt)   5.0853 7.5805  0.6708 0.5023 -9.7723 19.9429
#> exp(disp) 0.9662 0.0148 65.0838 0.0000  0.9371  0.9952
#> Call:
#> DeltaGeneric(object = object, def = list("exp(wt)"), alpha = 0.05,
#>     z = FALSE, df = 30)
#>            est     se      t df      p     2.5%   97.5%
#> exp(wt) 5.0853 7.5805 0.6708 30 0.5075 -10.3962 20.5668
#> Test passed with 1 success.

#> test-deltaMethod-delta

#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> Delta(coef = coef, vcov = vcov, func = func, alpha = 0.05)
#>              est     se       z      p    2.5%   97.5%
#> exp(wt)   5.0853 7.5805  0.6708 0.5023 -9.7723 19.9429
#> exp(disp) 0.9662 0.0148 65.0838 0.0000  0.9371  0.9952
#> Call:
#> Delta(coef = coef[2], vcov = vcov[2, 2, drop = FALSE], func = func,
#>     alpha = 0.05, z = FALSE, df = 30)
#>            est     se      t df      p     2.5%   97.5%
#> exp(wt) 5.0853 7.5805 0.6708 30 0.5075 -10.3962 20.5668
#> Test passed with 1 success.

#> test-zzz-coverage

#> Test passed with 3 successes.
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
#> function (program)
#> {
#>     testthat::skip_on_cran()
#>     testthat::skip_on_ci()
#>     testthat::skip_if(condition = Sys.which(names = program) ==
#>         "", message = paste0(program, " is not installed"))
#> }
#>
```

```
#> [[1]][[1]]$visible
#> [1] FALSE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
```

```
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
```

```
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
#> [[1]][[11]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[14]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[15]]
```

```
#> [[1]][[15]]$value
#> [[1]][[15]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[15]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[16]]
#> [[1]][[16]]$value
#> [[1]][[16]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[16]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[17]]
#> [[1]][[17]]$value
#> [[1]][[17]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[18]]
#> [[1]][[18]]$value
#> [[1]][[18]]$value[[1]]
#> NULL
#>
#>
#> [[1]][[18]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[19]]
#> [[1]][[19]]$value
#> [[1]][[19]]$value[[1]]
#> [1] TRUE
#>
#>
```

```
#> [[1]][[19]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[20]]
#> [[1]][[20]]$value
#> [[1]][[20]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[20]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[21]]
#> [[1]][[21]]$value
#> [[1]][[21]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[21]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[22]]
#> [[1]][[22]]$value
#> [[1]][[22]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[22]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[23]]
#> [[1]][[23]]$value
#> [[1]][[23]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[23]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[24]]
```

```
#> [[1]][[24]]$value
#> [[1]][[24]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[24]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[25]]
#> [[1]][[25]]$value
#> [[1]][[25]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[25]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[26]]
#> [[1]][[26]]$value
#> [[1]][[26]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[26]]$visible
#> [1] TRUE
```

Environment

```
ls()

#> [1] "nas1982" "root"
```

Class

```
#> [[1]]
#> [1] "data.frame"
#>
#> [[2]]
#> [1] "root_criterion"
```

References

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*, *58*(6), 1183–1186. https://doi.org/10.1080/00273171.2023.2201277

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/